

# Algebra II with Trigonometry

## Chapter 7.3

### Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:  
[4592201\\_alg-2trig-h-chp-7-3-note-docx.pdf](#)

#### Problems

1. Evaluate the product

$$32 \cos\left(\frac{\pi}{33}\right) \cos\left(\frac{2\pi}{33}\right) \cos\left(\frac{4\pi}{33}\right) \cos\left(\frac{8\pi}{33}\right) \cos\left(\frac{16\pi}{33}\right).$$

2. Find the exact value of  $\sin 12^\circ + \sin 48^\circ - \sin 72^\circ$ .

3. Evaluate  $\csc\left(\frac{\pi}{9}\right) \sqrt{2 - \sqrt{2 + \sqrt{2 + 2 \cos\left(\frac{8\pi}{9}\right)}}$ .

4. Find the exact value of  $\tan\left(2 \tan^{-1}(2) - \sin^{-1}\left(\frac{4}{5}\right)\right)$ .

5. Find the exact value of the sum  $\sum_{k=1}^6 \sin^4\left(\frac{(2k-1)\pi}{12}\right)$ .

6. Evaluate  $\frac{\sin 20^\circ + \sin 40^\circ + \sin 60^\circ + \sin 80^\circ + \sin 100^\circ}{\cos 20^\circ + \cos 40^\circ + \cos 60^\circ + \cos 80^\circ + \cos 100^\circ}$ .